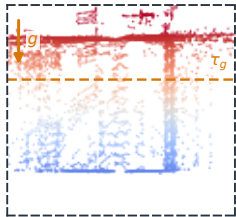
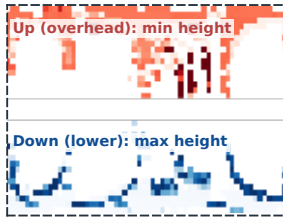


LIO deskew + inertial gravity → gravity canonicalization (roll/pitch removed, yaw free); shared map/query front end

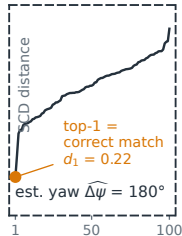
a Canonicalized query scan



b Dual-envelope SCD



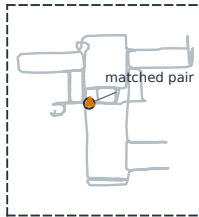
c Masked retrieval



d 6-DoF ICP verification



e Place association



side view of one measured scan

ring x sector (16 x 60), white = masked

candidate rank (top-100)

yaw-seeded overlay, before ICP

loop closure / relocalization